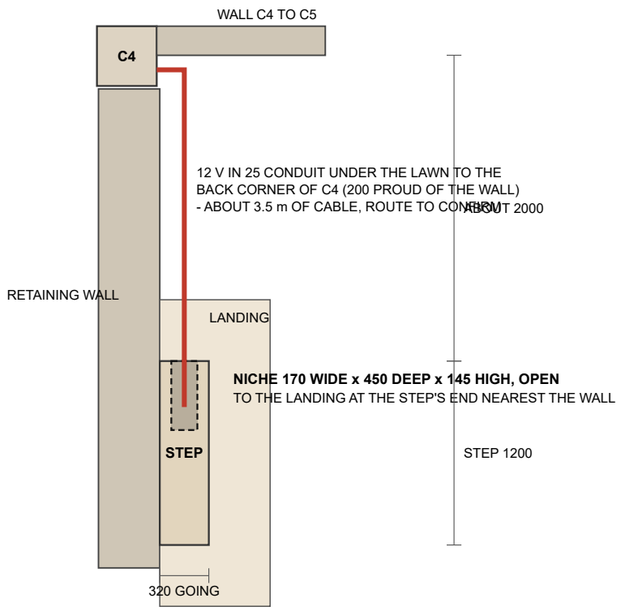
BLOCK WALL BW01 - SHEET 6: LED DRIVER FORMED INTO THE STEP BEHIND C4

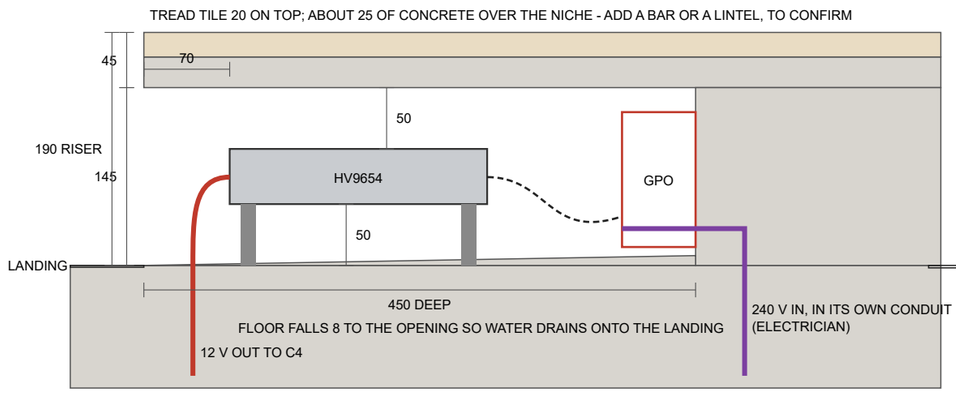
13 NOV 2026 · Havit HV9654, 100 W, 12 V, for the 14 Curva column lights and 2 step lights · niche at the end of the step nearest the wall

PRELIMINARY - FOR THE ELECTRICIAN TO CONFIRM BEFORE FORMING

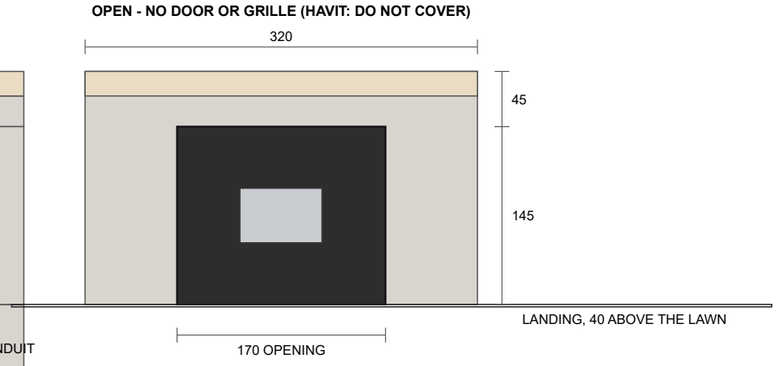
1. PLAN BEHIND C4, 1:40 (LOOKING DOWN)



2. SECTION ALONG THE NICHE (INTO THE STEP FROM ITS END), 1:5



3. THE STEP'S END FACE (FACING THE WALL), 1:5



WHY AN OPEN NICHE

Havit's installation sheet for the HV9654: the driver can become very hot and must not be covered; keep 50 mm clear between the driver and the surfaces round it; a covered driver voids the warranty. So the niche stays open to the landing, with no door or grille, and the driver sits on stand-offs 50 above the floor with 50 clear above and to each side.

WHY HERE

With the driver in the step and its 12 V fed into the existing 2.5 mm² at C4 (about 3.5 m of cable), the lowest light gets about 11.6 V and the longest run is about 17 m - inside Havit's limits (11.40 V or more, 25 m or less). With the driver at the GPO 18 m south of C1 every light failed both.

SIZES

Driver 210 long x 67.5 wide x 44.8 high (Havit data sheet). Niche 170 wide x 145 high x 450 deep: 50 + 67.5 + 50 across, 50 + 45 + 50 high, 70 in front of the driver, 60 to the outlet behind it, 60 for the outlet. Step 320 going x 190 riser; about 25 of concrete is left over the niche under a 20 tread tile.

NOTES - TO CONFIRM WITH THE ELECTRICIAN BEFORE THE STEP IS FORMED

1. 240 V: the driver comes with a 1 m flex and plug, and Havit says removing the plug voids the warranty, so the niche needs a weatherproof 240 V outlet the plug fits, drawn on the back wall of the niche. The electrician brings 240 V to it in its own conduit and chooses the outlet; if no outlet fits 145 high, the niche gets taller or the outlet goes in a pit beside the step.
2. 12 V out: the driver's 0.5 m output lead joins a twin 2.5 mm² in 25 conduit from the niche to C4, entering C4's core at the base, where it meets the existing 2.5 mm² in the footing (west to C1, east to C7). The joint must stay reachable (a draw box), not buried in core fill.
3. The 2.5 mm² already pulled from the GPO end to C1 is no longer needed for 12 V.
4. Earth: the HV9654 is Class 1 (earth required).
5. Over the niche: about 25 of concrete under the tread tile; add a bar or a small lintel when forming. Form the floor to fall 8 to the opening so water drains onto the landing.
6. Havit: installed by a licensed electrician to AS/NZS 3000; working temperature -25 to +50 C - an open niche on the shaded side of the step helps.